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VATSA: Video, Audio, Text, Sensory, Action

A Unified Five-Modality Architecture for

Human-Level Perception and Action

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Abstract

We present **VATSA** (Video, Audio, Text, Sensory, Action), a proposed unified architecture for human-level multimodal AI that integrates five distinct perceptual and actuation streams within a single coherent framework. While state-of-the-art multimodal models such as GPT-4o (OpenAI, 2024), Gemini Ultra, and Uni-MoE (Li et al., 2024) span two to four modalities, no existing system jointly addresses video, audio, text, physiological/IoT sensory data, and grounded action. Recent survey work on unified multimodal understanding (Yang et al., 2025) explicitly identifies the absence of sensory integration and closed-loop action as critical open frontiers.

VATSA addresses these gaps through four architectural principles: (1) a *shared latent space* in which all modality encoders project into a common high-dimensional embedding; (2) *cross-modal attention* enabling dynamic inter-modality interaction at the representation level; (3) a *temporal coherence layer* that synchronises streams with heterogeneous sampling rates; and (4) a *closed-loop action head* supporting physical, digital, and communicative outputs.

We present the conceptual architecture, motivating applications in healthcare, regulated pharmaceutical environments, autonomous systems, and adaptive education, an analysis of open research questions, and a phased implementation roadmap (2026–2028). This paper constitutes a timestamped declaration of the architectural hypothesis, providing a foundation for systematic empirical validation as each modality module is built and published openly. Benchmarks and experimental results will be incorporated in subsequent revisions.

Keywords: multimodal learning, unified architecture, cross-modal attention, sensory fusion, embodied AI, vision-language-action models

Contents

1 Introduction

Human intelligence is fundamentally multimodal. A clinician assessing a patient does not read a chart alone — they listen to the patient speak, observe facial expression and gait, monitor vital signs from sensors, and simultaneously formulate and execute a treatment plan. This seamless, context-aware fusion of multiple perceptual streams with grounded action is the hallmark of human-level cognition, and it remains largely unrealised in artificial systems.

Contemporary AI has made spectacular progress within individual modalities. Large language models (Vaswani et al., 2017) exhibit remarkable language understanding; vision-language models (Radford et al., 2021; Alayrac et al., 2022) fuse image and text; speech models handle audio; and vision-language-action (VLA) models (Ma et al., 2024) begin to integrate perception with robotic control. Yet even the most capable multimodal systems today address at most three or four modalities in a unified way, and none integrates physiological or IoT sensory streams alongside language, audio, video, and action within a single architecture.

VATSA (Video, Audio, Text, Sensory, Action) is a proposed architecture designed to fill this gap. It is motivated by a core claim: *five modalities, not two or three, are required to replicate the breadth of human sensorimotor intelligence*. The five streams are:

- **Video** — spatial and temporal scene understanding
- **Audio** — speech, sound events, prosody, environmental audio
- **Text** — language understanding, reasoning, instruction following
- **Sensory** — physiological signals, IoT sensors, proprioception
- **Action** — closed-loop physical or digital output

This paper presents the conceptual architecture of VATSA, its relationship to existing work, motivating application domains, open research questions, and the phased implementation roadmap through which it will be empirically tested. The repository hosting all associated code and documentation is publicly available at `https://github.com/vinaykumarkv/VATSA`.

Scope of this preprint. This is a concept and roadmap paper. It does not report experimental results; those will be incorporated as each phase of the roadmap is completed. The purpose of early publication is to establish a timestamped, peer-visible record of the architectural hypothesis and to invite community engagement, critique, and collaboration from the outset.

2 Background and Related Work

2.1 Multimodal Foundation Models

The past four years have seen a rapid convergence of modalities within foundation models. CLIP (Radford et al., 2021) demonstrated powerful vision-language alignment via contrastive learning. Flamingo (Alayrac et al., 2022) introduced few-shot visual language models. GPT-4o (OpenAI, 2024) extended this to audio, supporting real-time speech interaction alongside vision and text. Gemini Ultra similarly unifies three modalities. Uni-MoE (Li et al., 2024) employs Mixture-of-Experts (MoE) routing across four modalities (video, audio, image, text), representing the most comprehensive publicly described architecture to date.

Crucially, Yang et al. (2025) conducted a comprehensive survey of 700+ papers on unified multimodal understanding and generation, and explicitly identifies two outstanding frontiers: (1) the integration of physiological and IoT sensory data, and (2) action grounding within a unified architecture. VATSA addresses both simultaneously.

2.2 Shared Latent Space Modelling

ShaLa (Cui et al., 2025) proposes multimodal shared latent space modelling via diffusion, demonstrating that a common embedding space can support cross-modal generation and alignment beyond vision-language pairs. VATSA extends this principle across five modalities, treating the shared latent space as the central representational substrate.

2.3 Embodied AI and Vision-Language-Action Models

Ma et al. (2024) provide the first comprehensive taxonomy of vision-language-action (VLA) models, covering architectures that ground language and vision in physical robotic action. Liu et al. (2025) survey the broader landscape of embodied AI, identifying temporal alignment across heterogeneous sensing modalities as an open challenge. VATSA incorporates an action head directly into the unified architecture, extending VLA principles to include audio and sensory streams.

2.4 Generalist Agents

Gato (Reed et al., 2022) demonstrated that a single transformer can be trained across diverse tasks spanning text, images, and robotic control, establishing the viability of truly generalist architectures. VATSA builds on this spirit while focusing specifically on the five modalities most relevant to human-environment interaction.

2.5 The Modality Gap

Table 1 summarises the modality coverage of leading models. No existing published architecture covers all five VATSA modalities.

Table 1: Modality coverage of leading multimodal models. ✓ = supported; ∼ = partial; × = not supported.

Model	Video	Audio	Text	Sensory	Action	# Mod.
GPT-4o (OpenAI, 2024)	∼	✓	✓	×	×	3
Gemini Ultra (Google, 2023)	✓	✓	✓	×	×	3
LLaVA / LLaMA 3.2	✓	×	✓	×	×	2
Gato (Reed et al., 2022)	∼	×	✓	∼	✓	3–4
Uni-MoE (Li et al., 2024)	✓	✓	✓	×	×	4
AudioPaLM (Barrault et al., 2023)	×	✓	✓	×	×	2
VATSA (proposed)	✓	✓	✓	✓	✓	**5**

3 The VATSA Architecture

VATSA is built on four interconnected architectural principles. Figure 1 provides a schematic overview.

3.1 Principle 1: Shared Latent Space

Each modality encoder E_m ($m \in \{V, A, T, S\}$) projects its input x_m into a common high-dimensional embedding space $\mathcal{Z} \subset \mathbb{R}^d$:

$$z_m = E_m(x_m), \quad z_m \in \mathcal{Z}.$$

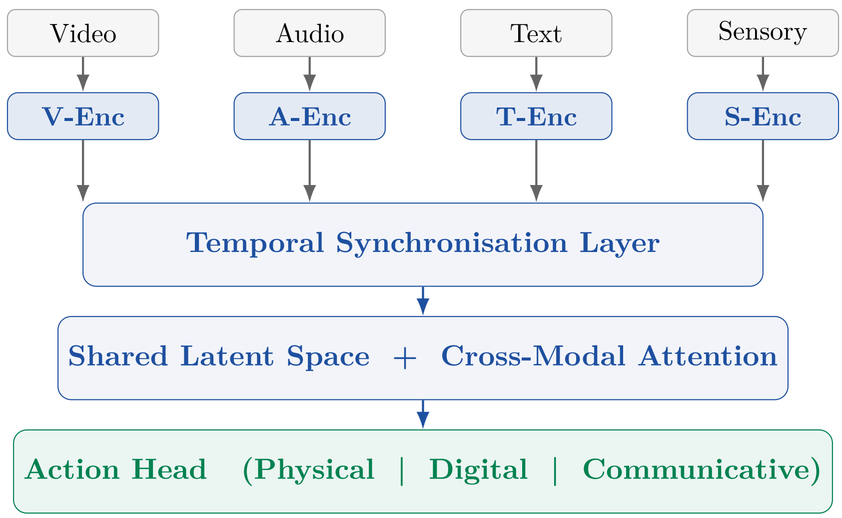

Figure 1: High-level schematic of the VATSA architecture. Each modality stream is independently encoded (Row 1), temporally aligned (Row 2), fused in a shared latent space via cross-modal attention (Row 3), and decoded through a multi-output action head (Row 4).

Representations from all five streams are thereby comparable, combinable, and jointly reasoned over. This extends the CLIP contrastive alignment principle (Radford et al., 2021) beyond vision-language pairs to a five-way shared space, and is informed by the shared latent space modelling approach of ShaLa (Cui et al., 2025).

3.2 Principle 2: Cross-Modal Attention

A central cross-modal transformer attends across all five modality streams simultaneously. Rather than early fusion (concatenating raw inputs) or late fusion (combining final predictions), VATSA employs *mid-fusion*: each modality is independently encoded, then fused at the representation level through multi-head cross-attention (Vaswani et al., 2017):

$$\mathrm{Attn}(Q_m, K_{m'}, V_{m'}) = \mathrm{softmax}\left(\frac{Q_m K_{m'}^{\top}}{\sqrt{d_k}}\right) V_{m'}$$

for all modality pairs (m, m'). Each modality influences and is influenced by the others dynamically, with attention weights learned end-to-end. The MoE routing strategy from Uni-MoE (Li et al., 2024) offers a promising efficient scaling path for this component.

3.3 Principle 3: Temporal Coherence

Video, audio, and sensory data are inherently temporal but have drastically different sampling rates: video at 25–60 fps, speech audio at 16–44 kHz, physiological sensors at 1–256 Hz. A temporal synchronisation layer aligns representations across modalities at matching time indices, addressing a challenge identified as open by Liu et al. (2025). This layer interpolates or aggregates modality embeddings to a shared temporal grid before cross-modal attention is applied.

3.4 Principle 4: Closed-Loop Action

VATSA includes an action head that produces outputs in three categories:

- **Physical:** motor control signals for robotic or prosthetic actuators
- **Digital:** API calls, form completions, system interactions
- **Communicative:** natural language generation

This closes the perception-action loop missing from most current multimodal models, and positions VATSA within the VLA paradigm (Ma et al., 2024) while extending action grounding to include sensory and audio conditioning.

4 Target Applications

VATSA's five-modality architecture enables applications that are infeasible or severely degraded with existing two- to four-modality systems.

4.1 Healthcare and Clinical Intelligence

Clinical decision support requires simultaneous access to structured measurements (Sensory), unstructured speech and patient-reported symptoms (Audio + Text), and physical observation (Video). VATSA can support:

- **Patient monitoring:** fusing vital signs with speech and observation for early deterioration detection
- **Surgical assistance:** real-time video plus sensory feedback with procedural knowledge for robotic surgical guidance
- **Mental health assessment:** prosody, language, and facial expression jointly modelled for holistic emotional state inference

4.2 Pharmaceutical and GxP-Regulated Environments

The first author has eight years of AI delivery experience in GxP-regulated pharmaceutical settings (GSK via TCS). Regulated manufacturing and quality control require multi-signal audit trails: sensor readings from equipment (S), video of operator procedures (V), verbal instructions and batch records (A + T), and automated corrective actions (A). VATSA's architecture is designed to support explainable multi-signal anomaly detection compatible with regulatory requirements (21 CFR Part 11, EU GMP Annex 11).

4.3 Autonomous Systems and Robotics

Embodied agents operating in the physical world must perceive across all relevant streams and respond with grounded actions. VATSA supports human-robot collaboration scenarios in which spoken instruction (A + T), physical context (V), workspace sensor feedback (S), and motor output (A) must be jointly modelled in real time.

4.4 Adaptive Education

Learner state modelling via facial expression (V), speech hesitation and prosody (A), and interaction patterns (T) enables real-time adaptation of educational content to individual engagement and comprehension states.

5 Open Research Questions

The VATSA architecture raises a set of research questions that will drive the empirical programme:

Q1. Dynamic modality weighting

How should modality contributions be balanced when streams are absent, corrupted, or of variable quality? ShaLa (Cui et al., 2025) provides early grounding for dynamic weighting via learned latent mixture.

Q2. Embedding dimensionality
What is the minimum shared embedding dimensionality d across five modalities that preserves modality-specific information without catastrophic forgetting?

Q3. Temporal alignment at scale
How to maintain coherent temporal alignment across modalities with sampling rates spanning five orders of magnitude? Liu et al. (2025) identify this as open.

Q4. Compute-efficient training
Can VATSA be trained without large-scale compute budgets via curriculum learning, modular pre-training, or parameter-efficient fine-tuning?

Q5. Evaluation benchmarks
What new benchmarks are needed to evaluate a five-modality system? Yang et al. (2025) identify the absence of appropriate benchmarks as a critical gap. We plan to propose VATSA-Bench as part of the experimental programme.

Q6. Sensory-conditioned action grounding
How should VLA-style action grounding incorporate physiological sensory feedback as a conditioning signal?

Q7. Ethics and regulation
What are the ethical and regulatory implications of simultaneous video, audio, and physiological perception, particularly in healthcare and GxP settings?

Q8. MoE scaling
Can Mixture-of-Experts routing (Li et al., 2024) provide an efficient scaling path for five-modality cross-attention?

6 Implementation Roadmap

VATSA will be implemented iteratively and publicly. Each phase produces a validated component before integration. All code is published at `https://github.com/vinaykumarkv/VATSA`.

Table 2: VATSA implementation roadmap.

Phase	Timeline	Technical Focus	Deliverable
1	Now – Q3 2026	CNN; object detection; ResNet, EfficientNet	V-module: visual encoder
2	Q3 – Q4 2026	Wav2Vec fine-tuning; speech datasets	A-module: audio encoder
3	Q4 2026 – Q1 2027	Transformer; LLM fine-tuning; RAG	T-module: text encoder
4	Q1 – Q2 2027	Time-series modelling; LSTM; IoT fusion	S-module: sensory encoder
5	Q2 – Q3 2027	Cross-modal attention; shared embedding	V+A+T tri-modal prototype
6	Q3 2027 – Q1 2028	Full integration; RL-based action layer	VATSA prototype + paper

The current paper (v1.0) establishes the architectural hypothesis. A second major revision incorporating benchmark results is planned for Q1 2028, coinciding with Phase 6 completion.

7 Limitations

We acknowledge several significant limitations of this preprint:

No experimental results. This paper presents a conceptual architecture and has not yet been implemented or benchmarked. All claims about architectural merit are hypothetical and will require empirical validation.

Single-author scope. The architecture has been developed by a single practitioner-researcher. Community critique, domain expert input, and collaborative stress-testing are explicitly solicited.

Compute constraints. Full-scale training of a five-modality architecture will require substantial compute resources. Phases 1–5 are designed to be executable with modest resources (single GPU), but Phase 6 integration may require access to compute infrastructure beyond current capacity.

Reference transparency. References to 2024/2025 literature were identified with the assistance of Grok to supplement independent reading. All cited works are genuine and their claims are represented accurately to the best of the author's knowledge.

8 Conclusion

We have presented VATSA, a unified five-modality AI architecture integrating video, audio, text, sensory, and action streams. The architecture is motivated by the gap in existing multimodal systems, which address at most four modalities and consistently exclude physiological sensory data and closed-loop action from unified treatment. VATSA is grounded in four architectural principles — shared latent space, cross-modal attention, temporal coherence, and closed-loop action — and connects to a strong foundation of recent work in multimodal learning, embodied AI, and shared representation.

The architecture is at concept and roadmap stage. The primary contribution of this preprint is the timestamped public articulation of the hypothesis, enabling community scrutiny and collaboration from the earliest stage of the research programme. We welcome engagement via the GitHub repository and invite researchers working in multimodal AI, embodied systems, and regulated AI in healthcare to contribute critiques, suggestions, and potential collaborations.

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